

MATH 211 Analysis III — Problem Session 2

Problem 1

Show that $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$.

Solution. Recall that every open interval (a, b) with $a, b \in \mathbb{R}$, $a < b$ contains a rational and an irrational number. Let q be a rational number and let $\epsilon > 0$ be any real number. The open ball around q with radius ϵ , $\text{Br}(q, \epsilon)$, contains an irrational number, which implies that $\mathbb{Q}$ is not an open set in $\mathbb{R}$. To show that $\mathbb{Q}$ is not closed in $\mathbb{R}$ one must show that $\mathbb{R} \setminus \mathbb{Q}$ is not open in $\mathbb{R}$, which follows by a similar argument to the one we used for the fact that $\mathbb{Q}$ is not open in $\mathbb{R}$.

Problem 2

Find the interior and the closure of each of the following sets:

$$(a, b), \quad (a, b], \quad [a, b]$$

where $a, b \in \mathbb{R}$, $a < b$.

Solution. Let $x \in (a, b)$. The open ball centered at x with radius $\epsilon = \min\{x - a, b - x\}$ is contained in (a, b) , which implies that the interior of (a, b) is (a, b) . We don't have to check points outside of (a, b) because a point in the interior of a set must be in the set itself.

We now investigate the boundary of (a, b) . An interior point can't be in the boundary, thus it is sufficient to check the points in the complement of (a, b) . The points a and b are in the boundary because for every $\epsilon > 0$, $\text{Br}(a, \epsilon)$ and $\text{Br}(b, \epsilon)$ contain both points of (a, b) and points of the complement of (a, b) , namely $(-\infty, a] \cup [b, +\infty)$. If $x \in (-\infty, a] \cup [b, +\infty)$ and $x > b$ then the open ball around x with radius $x - b$ does not intersect (a, b) . Thus any point greater than b can't be in the boundary. The case $x < a$ is similar. Thus, the boundary of (a, b) is $\{a, b\}$. Finally, since the closure of a set is the union of its interior and boundary we get the closure of (a, b) as $[a, b]$.

The other two cases, $(a, b]$ and $[a, b]$, can be done similarly.

Problem 3

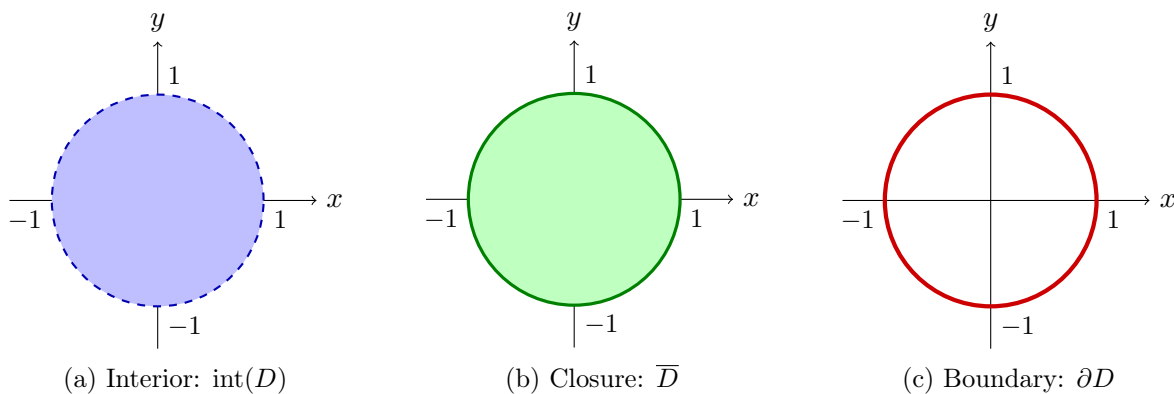
Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. Find the interior, the closure and the boundary of D . Sketch each of them and shade the corresponding regions.

Solution. We find the interior of D . Let x be a point in D . We claim $\text{Br}(x, 1 - \|x\|)$ is a subset of D . Indeed, if $y \in \text{Br}(x, 1 - \|x\|)$ then:

$$\|y\| = \|y - x + x\| \leq \|y - x\| + \|x\| < 1 - \|x\| + \|x\| = 1.$$

The first inequality is by triangle inequality and the second inequality holds since $y \in \text{Br}(x, 1 - \|x\|)$. Thus, D is an open set and the interior of D is D itself. For the boundary of D , for all the points x satisfying $\|x\| = 1$ the ball around x contains both points of D and points of the complement of D . Thus such points are in the boundary. For the points x satisfying $\|x\| > 1$ the ball around x with radius $\|x\| - 1$ does not intersect D (verify it). Thus, the boundary of D is only the points with norm 1. Thus, the closure of D is the set

$$\overline{D} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$



Problem 4

Show that the limit of a convergent sequence in $\mathbb{R}^n$ is unique.

Solution. Let x_k be a convergent sequence in $\mathbb{R}^n$. Suppose x and y are two limits of x_k . Let $\epsilon > 0$. By the definition of limit, there exist natural numbers N_1 and N_2 with the following properties:

- (i) $\|x_k - x\| < \epsilon$ for all $k > N_1$
- (ii) $\|x_k - y\| < \epsilon$ for all $k > N_2$.

Without loss of generality let $N_2 \geq N_1$. Then we have

$$\|x - y\| = \|x - y + x_k - x_k\| \leq \|x_k - y\| + \|x_k - x\| < 2\epsilon$$

for all $k > N_2$.

Since this holds for every $\epsilon > 0$, we get $\|x - y\| = 0$. By the properties of the norm function the only vector of zero norm is the zero vector. Thus $x - y = 0$, so $x = y$.

Problem 5

Prove that every Cauchy sequence in $\mathbb{R}^n$ converges.

Solution. In order to prove the statement we will employ the well-known theorem:

Theorem (Bolzano–Weierstrass). Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

The strategy of the proof is as follows: we first show that any Cauchy sequence x_k in $\mathbb{R}^n$ is bounded, then we conclude that this sequence has a convergent subsequence x_{k_j} (say converges to x) by using the Bolzano–Weierstrass theorem, and prove the theorem by showing x_k also converges to x .

Let x_k be a Cauchy sequence in $\mathbb{R}^n$. Let $\epsilon = 1$. By the definition of a Cauchy sequence there exists a natural number N such that $\|x_k - x_m\| < 1$ for every $k, m \geq N$. Thus $\|x_k\| \leq \|x_k - x_N\| + \|x_N\| < \|x_N\| + 1$ for every $k \geq N$. We managed to bound the terms of the sequence after the N 'th term but there might be finitely many points that are bigger than $\|x_N\| + 1$. We solve this issue by the following argument:

Let

$$M = \max\{\|x_1\|, \|x_2\|, \dots, \|x_{N-1}\|, \|x_N\| + 1\}.$$

It follows that $\|x_k\| \leq M$ for every k , thus x_k is a bounded sequence. Observe that we also proved that in order to show the boundedness of a sequence, we can ignore finitely many terms since we

can take the maximum over finitely many terms. Now by the Bolzano–Weierstrass theorem x_k has a subsequence x_{k_j} that converges (say to x). We prove x_k also converges to x .

Let $\epsilon > 0$. Then

$$\|x_k - x\| = \|x_k - x_{k_j} + x_{k_j} - x\| \leq \|x_k - x_{k_j}\| + \|x_{k_j} - x\| < \epsilon$$

for large enough k, j . $\|x_k - x_{k_j}\| < \frac{\epsilon}{2}$ as k, j go to $+\infty$ because x_k is a Cauchy sequence. $\|x_{k_j} - x\| < \frac{\epsilon}{2}$ because x_{k_j} converges to x .

Exercise. Show that every convergent sequence in $\mathbb{R}^n$ is a Cauchy sequence.

Remark. In this course, we only deal with the spaces in the form of $\mathbb{R}^n$. The notion of convergence of a sequence can be defined in a more general setting. The "spaces" (e.g. metric spaces, topological spaces, topological vector spaces of infinite dimensional) are called complete "spaces" if every Cauchy sequence in that "space" converges in that space.

Exercise. Find a sequence of rational numbers that does not converge to a rational number and conclude that $\mathbb{Q}$ is not complete.